

BUFFALO JOHNSON COUNTY AP, WY, USA

WMO: 726654

Lat: 44.381N

Lon: 106.721W

Elev: 4967

StdP: 12.24

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94054

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.2	-0.4	-13.6	3.2	-3.8	-8.2	4.2	3.6	36.3	24.4	32.6	24.2	8.2	350	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	92.8	60.3	89.9	59.9	86.4	59.3	64.4	82.5	63.0	81.8	61.7	80.8	10.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	87.9	70.8	56.6	82.1	69.8	54.6	76.4	68.7	32.1	82.1	30.9	81.1	29.8	80.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	27.3	24.1	DB	-13.8	98.5	5.4	2.8	-17.7	100.5	-20.8	102.2	-23.8	103.8	-27.8	105.8	(4)
(5)			WB	-14.3	67.9	5.3	3.8	-18.1	70.6	-21.2	72.8	-24.2	75.0	-28.1	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	28.1	26.5	36.3	43.5	51.8	61.9	72.2	70.0	60.2	46.7	36.2	27.0		
	DBStd	18.85	11.26	12.42	11.57	9.59	9.03	8.08	6.29	6.96	9.94	10.84	12.31	12.23		
	HDD50	3471	681	659	438	235	89	7	0	0	27	187	433	715		
	HDD65	7196	1145	1078	890	645	418	152	15	35	204	570	864	1180		
	CDD50	2307	0	1	13	40	144	363	689	621	331	86	18	1		
	CDD65	558	0	0	0	0	7	59	239	190	59	4	0	0		
	CDH74	6547	0	0	1	15	114	731	2746	2140	742	57	1	0		
	CDH80	2678	0	0	0	0	21	276	1229	895	249	8	0	0		
	WSAvg	9.4	8.6	9.0	10.0	11.4	10.8	9.4	8.7	8.7	9.1	9.4	9.0	8.7		
	PrecAvg	15.0	0.5	0.5	0.9	1.7	2.6	2.4	1.6	0.7	1.1	1.4	0.6	0.4		
	PrecMax	19.1	1.2	1.2	1.8	4.0	6.4	5.2	4.6	1.9	2.4	3.9	1.7	0.9		
(15) Precipitation	PrecMin	8.8	0.0	0.0	0.2	0.6	0.2	0.6	0.2	0.0	0.1	0.2	0.2	0.1		
	PrecStd	2.7	0.3	0.2	0.4	0.9	1.7	1.2	1.0	0.5	0.6	0.9	0.4	0.2		
	0.4% DB	56.6	59.6	70.2	76.6	82.5	93.5	98.2	96.0	90.7	80.7	70.2	57.8			
	0.4% MCWB	40.6	42.8	46.9	51.5	56.8	60.8	61.5	60.2	58.2	52.6	47.2	40.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	50.5	52.8	63.4	70.7	76.9	87.4	93.6	91.5	86.4	73.8	63.5	52.0			
	2% MCWB	37.6	39.4	44.5	48.8	55.5	60.3	61.1	59.2	56.4	50.2	44.3	38.2			
	5% DB	47.2	47.9	58.6	65.2	72.5	82.1	90.4	88.9	82.4	69.0	58.1	47.1			
	5% MCWB	35.9	36.6	42.3	46.9	54.3	59.8	60.8	59.2	55.3	48.9	41.9	35.6			
	10% DB	43.5	43.4	53.5	60.7	67.9	77.3	87.4	85.0	78.2	63.5	53.7	43.2			
	10% MCWB	33.6	33.8	40.4	45.0	52.4	59.0	60.7	58.4	53.9	46.4	39.8	33.6			
	0.4% WB	41.4	43.3	48.4	53.2	60.2	66.4	66.6	64.5	61.6	54.7	48.3	41.9			
	0.4% MCDB	54.5	57.5	67.3	73.6	75.1	80.6	85.0	83.4	80.7	73.0	67.8	56.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.5	39.9	45.3	49.9	57.4	63.6	65.0	63.0	58.9	52.0	45.1	38.7			
	2% MCDB	49.4	51.4	61.8	67.1	72.3	79.8	83.3	82.9	79.5	70.2	62.1	50.1			
	5% WB	36.3	36.9	42.9	47.7	55.5	61.7	63.6	61.5	56.9	49.8	42.7	36.2			
	5% MCDB	46.1	47.1	57.1	63.2	70.1	78.6	82.5	81.1	77.8	66.4	57.4	46.3			
	10% WB	34.1	34.3	40.7	45.6	53.4	59.5	62.2	60.1	55.0	47.5	40.3	33.8			
	10% MCDB	42.8	42.5	52.4	59.3	66.2	75.2	81.8	79.8	74.7	62.3	52.4	42.6			
(35) Mean Daily Temperature Range	MDBR	18.3	17.9	20.0	20.9	20.3	22.7	25.5	25.1	23.7	20.9	19.3	17.7			
	5% DB	23.0	22.7	27.3	29.3	27.1	29.5	29.7	29.2	29.2	28.4	25.2	22.3			
	5% MCWBR	14.2	13.4	14.1	14.3	13.1	12.7	10.7	10.6	11.7	13.4	13.5	13.5			
	5% WB	22.4	21.5	25.5	27.0	24.7	26.5	26.1	25.8	26.2	26.0	23.9	21.2			
	5% MCWBR	14.2	13.3	13.5	13.8	12.9	13.0	10.8	10.1	11.0	12.9	13.2	13.3			
(40) Clear-Sky Solar Irradiance	taub	0.238	0.247	0.268	0.292	0.314	0.321	0.342	0.341	0.308	0.280	0.251	0.239			
	taud	2.559	2.550	2.542	2.487	2.489	2.494	2.465	2.445	2.533	2.578	2.570	2.551			
	Ebn at Noon	293	308	312	309	303	300	292	289	293	290	284	281			
	Edh at Noon	20	24	28	32	33	33	34	33	28	23	19	18			
(44) All-Sky Solar Radiation	RadAvg	590	886	1276	1597	1876	2239	2279	1950	1530	993	651	489			
	RadStd	32	46	89	97	186	162	87	75	111	82	51	31			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page